

Sol. $\sqrt{1 + \frac{x}{144}} = \frac{13}{12} \Rightarrow \left(1 + \frac{x}{144}\right) = \left(\frac{13}{12}\right)^2 = \frac{169}{144} \Rightarrow \frac{x}{144} = \frac{169}{144} - 1$
 $\Rightarrow \frac{x}{144} = \frac{25}{144} \Rightarrow x = 25.$

Ex. 13. Simplify : $\frac{1}{\sqrt{100} - \sqrt{99}} - \frac{1}{\sqrt{99} - \sqrt{98}} + \frac{1}{\sqrt{98} - \sqrt{97}} - \frac{1}{\sqrt{97} - \sqrt{96}} + \dots + \frac{1}{\sqrt{2} - \sqrt{1}}.$ **(Section Officers', 2005)**

Sol. Given expression

$$= \frac{1}{\sqrt{100} - \sqrt{99}} \times \frac{\sqrt{100} + \sqrt{99}}{\sqrt{100} + \sqrt{99}} - \frac{1}{\sqrt{99} - \sqrt{98}} \times \frac{\sqrt{99} + \sqrt{98}}{\sqrt{99} + \sqrt{98}} + \frac{1}{\sqrt{98} - \sqrt{97}} \times \frac{\sqrt{98} + \sqrt{97}}{\sqrt{98} + \sqrt{97}}$$

$$- \frac{1}{\sqrt{97} - \sqrt{96}} \times \frac{\sqrt{97} + \sqrt{96}}{\sqrt{97} + \sqrt{96}} + \dots + \frac{1}{\sqrt{2} - \sqrt{1}} \times \frac{\sqrt{2} + \sqrt{1}}{\sqrt{2} + \sqrt{1}}$$

$$= \frac{\sqrt{100} + \sqrt{99}}{(100 - 99)} - \frac{\sqrt{99} + \sqrt{98}}{(99 - 98)} + \frac{\sqrt{98} + \sqrt{97}}{(98 - 97)} - \frac{\sqrt{97} + \sqrt{96}}{(97 - 96)} + \dots + \frac{\sqrt{2} + \sqrt{1}}{(2 - 1)}$$

$$= (\sqrt{100} + \sqrt{99}) - (\sqrt{99} + \sqrt{98}) + (\sqrt{98} + \sqrt{97}) - (\sqrt{97} + \sqrt{96}) + \dots + (\sqrt{2} + \sqrt{1})$$

$$= \sqrt{100} + \sqrt{1} = 10 + 1 = 11.$$

Ex. 14. Find the sum : $3 + \frac{1}{\sqrt{3}} + \frac{1}{3 + \sqrt{3}} - \frac{1}{3 - \sqrt{3}}.$ **(S.S.C., 2007)**

Sol. $3 + \frac{1}{\sqrt{3}} + \frac{1}{3 + \sqrt{3}} - \frac{1}{3 - \sqrt{3}} = 3 + \frac{1}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} + \frac{1}{3 + \sqrt{3}} \times \frac{3 - \sqrt{3}}{3 - \sqrt{3}} - \frac{1}{3 - \sqrt{3}} \times \frac{3 + \sqrt{3}}{3 + \sqrt{3}}$
 $= 3 + \frac{\sqrt{3}}{3} + \frac{3 - \sqrt{3}}{(9 - 3)} - \frac{3 + \sqrt{3}}{(9 - 3)} = 3 + \frac{\sqrt{3}}{3} + \frac{3 - \sqrt{3}}{6} - \frac{3 + \sqrt{3}}{6}$
 $= \frac{18 + 2\sqrt{3} + 3 - \sqrt{3} - 3 - \sqrt{3}}{6} = \frac{18}{6} = 3.$

Ex. 15. Find the value of $\sqrt{3}$ upto three places of decimal.

Sol.

1	3.000000 (1.732
	1
27	200
	189
343	1100
	1029
3462	7100
	6924

$\therefore \sqrt{3} = 1.732.$

Ex. 16. If $\sqrt{3} = 1.732$, find the value of $\sqrt{192} - \frac{1}{2}\sqrt{48} - \sqrt{75}$ correct to 3 places of decimal. **(S.S.C., 2004)**

Sol. $\sqrt{192} - \frac{1}{2}\sqrt{48} - \sqrt{75} = \sqrt{64 \times 3} - \frac{1}{2}\sqrt{16 \times 3} - \sqrt{25 \times 3} = 8\sqrt{3} - \frac{1}{2} \times 4\sqrt{3} - 5\sqrt{3}$
 $= 3\sqrt{3} - 2\sqrt{3} = \sqrt{3} = 1.732.$

Ex. 17. If $\sqrt{0.05 \times 0.5 \times a} = 0.5 \times 0.05 \times \sqrt{b}$, then find the value of $\frac{a}{b}$. **(M.B.A., 2006)**

Sol. Clearly, we have: $\frac{\sqrt{a}}{\sqrt{b}} = \frac{0.5 \times 0.05}{\sqrt{0.05 \times 0.5}} \Rightarrow \frac{a}{b} = \frac{0.5 \times 0.05 \times 0.5 \times 0.05}{0.05 \times 0.5} = 0.5 \times 0.05 = 0.025.$